Student Performance Analytics System

1. Project Title

Student Performance Analytics System

2. Objective

The main objective of this project is to analyze student academic performance using Python. It helps to understand students' marks, grades, attendance, and overall results in an organized way.

Instead of calculating everything manually, this project uses Python to perform calculations and display useful information about student performance. It also helps identify students who performed well and subjects in which students may need improvement.

3. Technologies Used

The following technologies and libraries are used in this project:

- Python: Used to write the program and implement the calculations.

- Pandas: Used to read the CSV file, organize student data, and perform data analysis.

- NumPy: Used for numerical calculations where required.

- CSV: Used to store the student dataset.

- Visual Studio Code: Used to write and run the Python program.

4. Dataset Description:

The dataset is stored in a file named "Students.csv". It contains records of 20 students from different branches.

The dataset includes student registration numbers, names, branches, year of study, attendance percentage, and marks in five subjects: Python, DBMS, Java, Operating Systems, and Computer Networks.

The program processes this information to calculate total marks, grade points, SGPA, and the final result of each student.

5. Implementation

First, the student dataset is loaded from the CSV file using Pandas and stored in a DataFrame. This makes it easier to access and process the information.Next, the program

calculates the total marks obtained by each student. The subject marks are then converted into grade points according to the grading rules used in the project.

A separate calculation is performed to find the total grade points and SGPA of each student. The SGPA is calculated by taking the average of the five subject grade points, assuming all subjects have equal credits.

The program then checks each student's subject marks to determine whether the student has passed or failed. It also identifies absent or missing subject values according to the implemented conditions.

After calculating the individual results, the program performs class-level analysis. It calculates the number of students who passed, failed, or were absent, along with the class average marks.Finally, the program performs subject-wise analysis, identifies the highest and lowest performers, displays the top five students, and saves the processed data in a report CSV file.

6. Key Features

The main features of the project are:

1. Reads student data from a CSV file.

2. Organizes data using a Pandas DataFrame.

3. Calculates total marks for each student.

4. Converts subject marks into grade points.

5. Calculates total grade points and SGPA.

6. Determines the pass or fail status of students.

7. Calculates class performance statistics.

8. Finds the average performance of each subject.

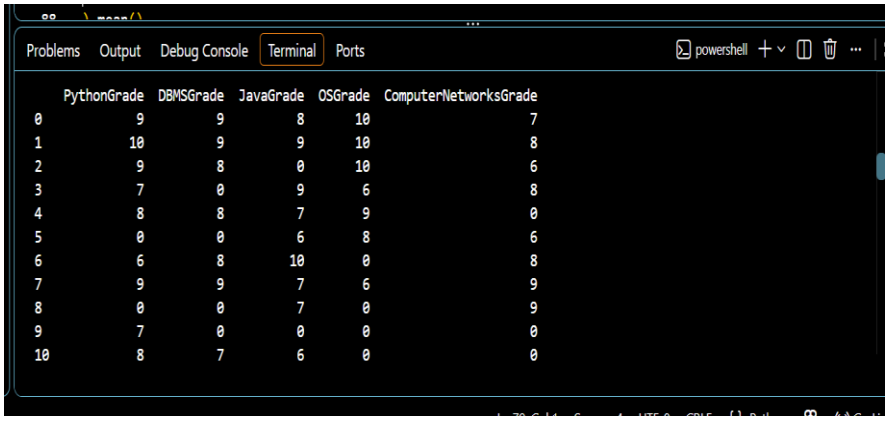
9. Identifies the best-performing and lowest-performing subjects.

10. Displays the top five students based on SGPA.

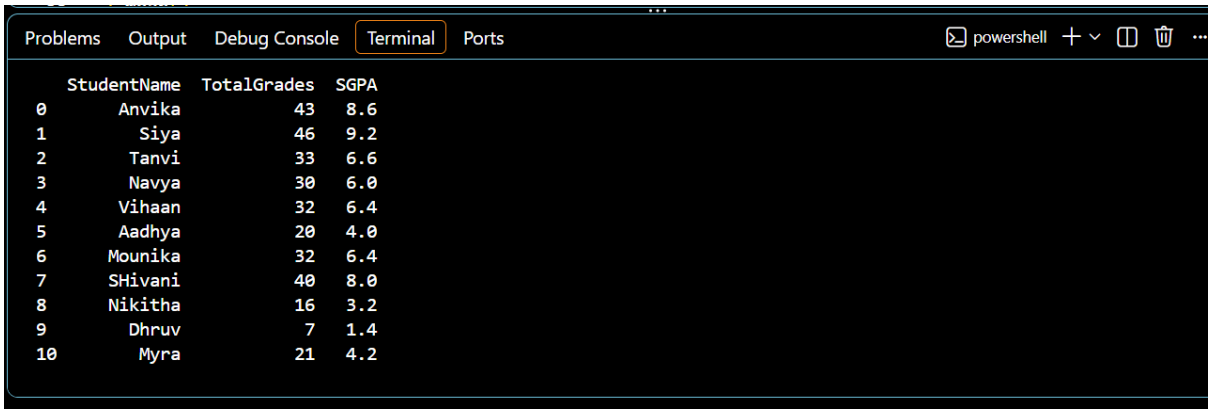
11. Saves the final processed data in a CSV report.

7. Output and Results

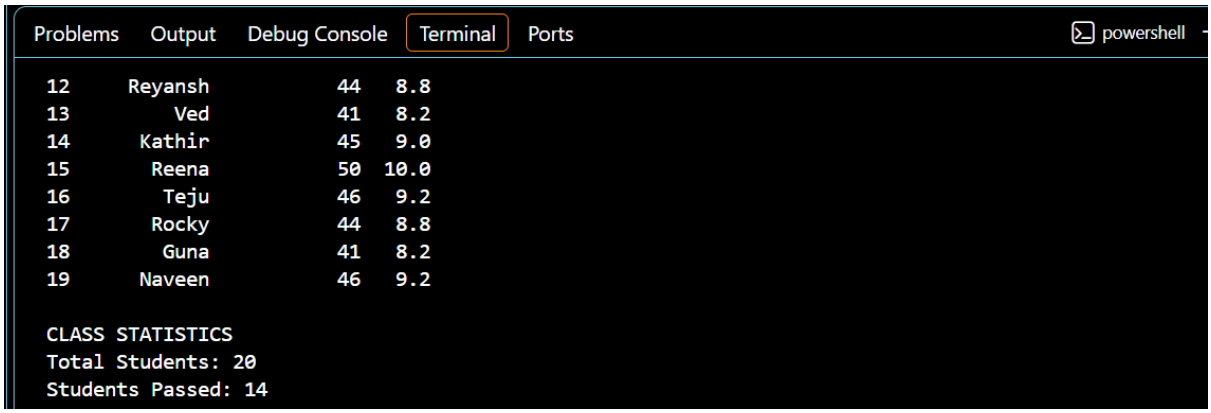
The program processes the records of 20 students and displays the results in the terminal as follows:



	PythonGrade	DBMSGrade	JavaGrade	OSGrade	ComputerNetworksGrade
0	9	9	8	10	7
1	10	9	9	10	8
2	9	8	0	10	6
3	7	0	9	6	8
4	8	8	7	9	0
5	0	0	6	8	6
6	6	8	10	0	8
7	9	9	7	6	9
8	0	0	7	0	9
9	7	0	0	0	0
10	8	7	6	0	0

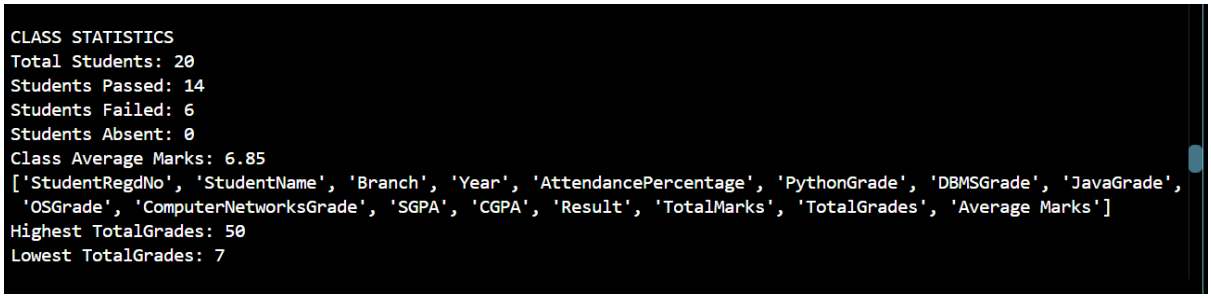


	StudentName	TotalGrades	SGPA
0	Anvika	43	8.6
1	Siya	46	9.2
2	Tanvi	33	6.6
3	Navya	30	6.0
4	Vihaan	32	6.4
5	Aadhya	20	4.0
6	Mounika	32	6.4
7	SHivani	40	8.0
8	Nikitha	16	3.2
9	Dhruv	7	1.4
10	Myra	21	4.2



12	Reyansh	44	8.8
13	Ved	41	8.2
14	Kathir	45	9.0
15	Reena	50	10.0
16	Teju	46	9.2
17	Rocky	44	8.8
18	Guna	41	8.2
19	Naveen	46	9.2

```
CLASS STATISTICS
Total Students: 20
Students Passed: 14
```



```
CLASS STATISTICS
Total Students: 20
Students Passed: 14
Students Failed: 6
Students Absent: 0
Class Average Marks: 6.85
['StudentRegdNo', 'StudentName', 'Branch', 'Year', 'AttendancePercentage', 'PythonGrade', 'DBMSGrade', 'JavaGrade',
 'OSGrade', 'ComputerNetworksGrade', 'SGPA', 'CGPA', 'Result', 'TotalMarks', 'TotalGrades', 'Average Marks']
Highest TotalGrades: 50
Lowest TotalGrades: 7
```

Problems Output Debug Console Terminal Ports powe

```
========== SUBJECT-WISE ANALYSIS ==========
Average Marks in Each Subject:
PythonGrade             6.80
DBMSGrade               6.60
JavaGrade               7.20
OSGrade                 7.15
ComputerNetworksGrade   6.50
dtype: float64

Best Performing Subject:
JavaGrade
```

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========== TOP 5 STUDENTS ==========
StudentName  TotalGrades  SGPA Result
      Reena           50  10.0   Pass
      Arjun           49   9.8   Pass
       Siya           46   9.2   Pass
       Teju           46   9.2   Pass
     Naveen           46   9.2   Pass

========== FINAL STUDENT REPORT ==========
StudentRegdNo StudentName Branch  Year  AttendancePercentage  PythonGrade  DBMSG
etworksGrade  SGPA  CGPA Result  TotalMarks  TotalGrades  Average Marks
   24331A1221      Anvika     IT     3                 91.5           8
```

Welcome | Students Data.csv | main.py | student_performance_report.csv

student_performance_report.csv

```
1   StudentRegdNo,StudentName,Branch,Year,AttendancePercentage,PythonGrade,DBMSGrade,Java
2   24331A1221,Anvika,IT,3,91.5,8,8,7,9,6,8.6,,Pass,38,43,7.6
3   24331A1222,Siya,IT,3,89.0,9,8,8,9,7,9.2,,Pass,41,46,8.2
4   24331A1223,Tanvi,IT,3,67.0,8,7,4,9,5,6.6,,Pass,33,33,6.6
5   24331A1224,Navya,IT,3,89.0,6,4,8,5,7,6.0,,Pass,30,30,6.0
6   24331A1225,Vihaan,IT,3,45.0,7,7,6,8,3,6.4,,Fail,31,32,6.2
7   24331A02B1,Aadhya,CSE,3,77.0,3,2,5,7,5,4.0,,Fail,22,20,4.4
8   24331A02B2,Mounika,CSE,3,90.0,5,7,9,3,7,6.4,,Fail,31,32,6.2
9   24331A02B5,SHivani,CSE,3,88.0,8,8,6,5,8,8.0,,Pass,35,40,7.0
10  24331A02B6,Nikitha,CSE,3,80.0,1,2,6,4,8,3.2,,Fail,21,16,4.2
11  24331A02B4,Dhruv,CSE,3,45.0,6,3,4,4,3,1.4,,Fail,20,7,4.0
12  24331A5201,Myra,ECE,3,78.0,7,6,5,4,0,4.2,,Fail,22,21,4.4
13  24331A5202,Arjun,ECE,3,90.0,9,9,9,8,9,9.8,,Pass,44,49,8.8
14  24331A5203,Reyansh,ECE,3,89.0,8,6,9,9,7,8.8,,Pass,39,44,7.8
15  24331A5204,Ved,ECE,3,56.0,7,6,8,9,6,8.2,,Pass,36,41,7.2
16  24331A5205,Kathir,ECE,3,89.0,7,7,9,9,8,9.0,,Pass,40,45,8.0
17  24331C1301,Reena,AIML,3,95.0,9,9,9,9,9,10.0,,Pass,45,50,9.0
18  24331C1302,Teju,AIML,3,89.0,7,9,8,8,9,9.2,,Pass,41,46,8.2
19  24331C1303,Rocky,AIML,3,85.0,7,8,8,9,7,8.8,,Pass,39,44,7.8
20  24331C1304,Guna,AIML,3,89.0,6,8,8,6,8,8.2,,Pass,36,41,7.2
21  24331C1305,Naveen,AIML,3,78.0,8,8,8,9,8,9.2,,Pass,41,46,8.2
22
```

During testing, the program showed the following class statistics:

- Total students: 20

- Students passed: 14

- Students failed: 6

- Students absent: 0

- Class average marks: 6.85

The program also performs subject-wise analysis and displays the top five students based on their SGPA.

The processed student information can be saved in "student_performance_report.csv" for further reference.

The statistics above are based on the program output observed during testing. The remaining outputs should be verified after the program runs successfully.

8. Final Outcome

Through this project, I learned how Python can be used to process and analyze student data. The project reduces manual calculations and makes it easier to understand student performance.I also learned how to work with CSV files, Pandas DataFrames, functions, loops, conditions, and numerical calculations. The project helped me understand how raw data can be converted into meaningful information.

9. Challenges and Learning

While developing this project, I faced some errors related to column names and calculations. For example, a mismatch between a DataFrame column name and the name used in the code caused a KeyError.

I learned that column names must match exactly when accessing data using Pandas. I also learned how to use functions for repeated calculations and how to debug errors by checking the program output.

This project improved my understanding of Python programming and basic data analysis.

10. GitHub Repository:

GitHub Repository Link: Upload files · Gayatri1520/Python-With-AI

The repository will contain the Python program, student dataset, README file, and project documentation.